🏥 HEALTHCARE PATIENT READMISSION ANALYSIS DASHBOARD

📊 End-to-End Data Analytics Project using Python, SQL & Power BI

🌟 Project Overview

The **Healthcare Patient Readmission Analysis Dashboard** is an end-to-end data analytics project designed to analyze hospital patient records, identify readmission trends, monitor treatment costs, and evaluate patient stay patterns across departments.

This project combines:

✅ **Python for EDA**
✅ **SQL for Data Analysis**
✅ **Power BI for Dashboard Development**

The main objective of this project is to transform raw healthcare data into meaningful business insights that help healthcare organizations improve patient care and operational efficiency.

🎯 Problem Statement

Hospital readmissions are one of the biggest challenges in the healthcare industry. Frequent readmissions increase treatment costs, create operational pressure, and indicate possible inefficiencies in patient care.

Healthcare organizations require a centralized analytics dashboard to:

🔹 Monitor patient readmission trends
🔹 Analyze treatment costs
🔹 Track department-wise performance
🔹 Understand patient demographics
🔹 Improve healthcare decision-making

This project aims to solve these problems through interactive analytics and visualization.

🚀 Project Workflow

Raw Dataset
↓
Data Cleaning (Python)
↓
Exploratory Data Analysis (EDA)
↓
SQL Analysis & Querying
↓
Power BI Dashboard Development
↓
Business Insights Generation

🛠️ Tools & Technologies Used

Tool / Technology	Purpose
🐍 Python	Data Cleaning & EDA
📘 Pandas	Data Manipulation
🔢 NumPy	Numerical Operations
📈 Matplotlib	Data Visualization
🎨 Seaborn	Statistical Charts
🛢️ SQL	Data Querying & Analysis
📊 Power BI	Dashboard Development
⚡ DAX	Measures & Calculations
🔄 Power Query	Data Transformation
📄 CSV / Excel	Dataset Storage

📂 Dataset Information

The healthcare dataset contains patient-related hospital information including:

- 🆔 Patient ID
- 👨‍⚕️ Age & Gender
- 🏥 Department
- 💉 Disease
- 📅 Admission & Discharge Date
- 💰 Treatment Cost
- 🔁 Readmission Status
- 🛏️ Length of Stay

🧹 Phase 1: Data Cleaning & Preprocessing

Before analysis, the raw dataset was cleaned and transformed using Python.

✅ Data Cleaning Steps

✔ Removed missing/null values
✔ Removed duplicate records
✔ Standardized column formats
✔ Converted date columns into datetime format
✔ Corrected inconsistent values
✔ Created calculated columns for better analysis

🧠 Feature Engineering

New columns were created for advanced analysis:

New Column	Purpose
Age Group	Categorize patients by age
Stay Category	Short / Medium / Long Stay
Cost Category	Low / Medium / High Cost

📊 Phase 2: Exploratory Data Analysis (EDA)

EDA was performed using Python to understand hidden patterns and healthcare trends.

🔍 Analysis Performed

📌 Patient Distribution Analysis

- Department-wise patient count
- Gender-wise patient analysis
- Age group distribution

📌 Readmission Analysis

- Monthly readmission trends
- High-risk patient groups
- Readmission percentages

📌 Treatment Cost Analysis

- Average treatment cost
- Cost category distribution
- High-cost treatment identification

📌 Length of Stay Analysis

- Average hospital stay
- Department-wise stay comparison

🐍 Python Libraries Used

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

🛢️ Phase 3: SQL Analysis

SQL was used to perform advanced data querying and healthcare analysis.

✅ SQL Operations Used

✔ SELECT Statements
✔ WHERE Conditions
✔ GROUP BY Analysis
✔ ORDER BY Sorting
✔ Aggregate Functions
✔ CASE Statements
✔ Subqueries

📌 SQL Business Analysis

🔹 Department-wise Patient Count

Identified departments with the highest patient admissions.

🔹 Readmission Rate Analysis

Analyzed patient readmission percentages.

🔹 Treatment Cost Analysis

Compared average treatment costs across departments.

🔹 Gender & Age Analysis

Studied demographic patterns in hospital admissions.

📈 Phase 4: Power BI Dashboard Development

An interactive healthcare dashboard was created in Power BI to visualize hospital performance and patient insights.

🎨 Dashboard Features

🧾 KPI Cards

The dashboard contains important healthcare metrics:

✅ Total Patients
✅ Readmitted Patients
✅ Non-Readmitted Patients
✅ Average Length of Stay
✅ Average Treatment Cost
✅ Total Treatment Cost

📊 Dashboard Visualizations

📈 Readmission Rate Over Time

- Displays monthly patient readmission trends.
- Helps monitor healthcare performance over time.

👥 Readmission Rate by Age Group

- Compares readmission percentages among different age groups.

🛏️ Average Length of Stay

- Gauge chart showing overall patient stay duration.

🏥 Total Patients by Department

- Horizontal bar chart displaying department-wise patient count.

🧩 Average Length of Stay by Department

- Treemap visualization comparing hospital stay duration across departments.

💰 Treatment Cost Analysis

- Donut chart showing treatment cost category distribution.

🚻 Patient ID by Gender

- Displays gender-wise patient distribution.

💡 Key Insights Section

- Summarizes important business findings from the dashboard.

⚡ Power Query Transformations

Power Query was used for:

✔ Data type conversion
✔ Conditional column creation
✔ Date formatting
✔ Cost category creation
✔ Stay category transformation

🧮 DAX Measures Created

📌 Total Patients

Total Patients = COUNT(PatientID)

📌 Readmitted Patients

```
Readmitted Patients =
CALCULATE(
  COUNT(PatientID),
  cleaned_hospital_data[Readmitted] = "Yes"
)
```

📌 Average Length of Stay

```
Average Length Of Stay =
AVERAGE(cleaned_hospital_data[LengthOfStay])
```

📌 Average Treatment Cost

```
Average Treatment Cost =
AVERAGE(cleaned_hospital_data[TreatmentCost])
```

🔥 Key Dashboard Insights

📌 Major Findings

- 🔹 Gastroenterology recorded the highest patient admissions.
- 🔹 General Medicine had the lowest patient count among major departments.
- 🔹 Readmission rates remained consistently high across multiple months.
- 🔹 High-cost treatments contributed significantly to total healthcare expenses.
- 🔹 Middle-aged and senior patients showed higher readmission rates.
- 🔹 Average patient stay duration remained above 7 days.
- 🔹 Female patients accounted for a significant portion of admissions.

📉 Business Impact

This dashboard helps healthcare organizations:

✅ Improve patient monitoring
✅ Track hospital performance
✅ Reduce operational inefficiencies
✅ Analyze treatment expenses
✅ Identify high-risk patient groups
✅ Support data-driven decision making

⚠️ Challenges Faced During the Project

- 🔸 Handling inconsistent healthcare data
- 🔸 Creating meaningful calculated columns
- 🔸 Designing a clean dashboard layout
- 🔸 Choosing premium color combinations
- 🔸 Managing visual alignment & responsiveness
- 🔸 Creating business-friendly insights

🎓 Key Learnings From This Project

Through this project, I gained practical experience in:

✨ Real-world healthcare data analytics
✨ Python-based data preprocessing
✨ SQL business analysis
✨ Dashboard storytelling
✨ DAX calculations
✨ Power BI design principles
✨ Business insight generation

🏁 Conclusion

The **Healthcare Patient Readmission Dashboard** successfully transformed raw hospital data into meaningful business insights using Python, SQL, and Power BI.

This project demonstrates the complete lifecycle of a Data Analytics project:

📌 Data Cleaning
📌 Exploratory Data Analysis
📌 SQL Querying
📌 Dashboard Development
📌 Business Insight Generation

The dashboard enables healthcare professionals to better understand patient behavior, treatment costs, and readmission trends for improved healthcare management.

🌟 Project Highlights

✅ End-to-End Analytics Project
✅ Real-World Healthcare Dataset
✅ Python + SQL + Power BI Integration
✅ Interactive Dashboard Design
✅ Premium UI/UX Dashboard Layout
✅ Business-Oriented Insights
✅ Data Cleaning & Feature Engineering
✅ KPI & DAX Measure Creation

🔗 Project Files Included

📁 Python EDA Notebook
📁 SQL Query File
📁 Cleaned Dataset
📁 Power BI Dashboard (.pbix)
📁 Dashboard Screenshots
📁 Project Documentation PDF

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⭐ If you liked this project, don't forget to give it a star on GitHub!